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Polish present active participles as syntactic centaurs

Abstract

This paper investigates the syntax of present active participles in Polish, morphologically similar forms that nevertheless split into agreeing and invariant types. Agreeing participles show full ϕ -agreement and appear in adjectival environments, while invariant participles lack agreement and function as adverbial, converb-like modifiers. Despite this contrast, both types retain core verbal properties, including argument structure and aspectual interpretation.

I argue that this split does not reflect a categorial distinction between verbs and adjectives. Instead, both participle types are derived from the same verbal structure and the difference follows from the presence or absence of an additional ϕ -probe and from structural position. Agreement is licensed by syntactic Agree, not by recategorization to adjective. Adopting a feature-driven minimalist approach, I propose that non-categorial features such as ϕ can provide labels under the Labeling Algorithm, making an adjectival head unnecessary. Present active participles will therefore be defined as ‘centaur’ constructions, showing properties of multiple categories, which, however, does not entail syntactic category change.

Keywords

active participles, converbs, feature-driven syntax, Labeling Algorithm, Minimalism, Polish, recategorization, ϕ -agreement.

Abstrakt

Przedmiotem artykułu jest analiza składni imiesłowów w języku polskim – form morfologicznie podobnych, wśród których można jednak wyróżnić typy odmienne (przymiotnikowe) i nieodmienne (przysłówkowe). Imiesłowy uzgadniane wykazują pełne

uzgodnienie cech ϕ i dystrybucję przymiotnikową, podczas gdy imiesłowy nieodmienne funkcjonują jako przysłówkowe modyfikatory o charakterze konwerbiów. Mimo tego kontrastu oba typy zachowują kluczowe właściwości werbalne, w tym strukturę argumentową oraz interpretację aspektową.

Dowodzę, że podział ten nie wynika z dystynkcji kategorialnej między czasownikiem a przymiotnikiem. Zamiast tego oba typy imiesłówów derywowane są z tej samej struktury werbalnej, a różnica wynika z obecności lub braku dodatkowej sondy ϕ oraz z pozycji strukturalnej. Uzgodnienie jest licencjonowane przez operację składniową Agree, a nie przez rekategoryzację do przymiotnika. Przyjmując oparte na cechach podejście minimalistyczne, proponuję rozwiązanie, w którym cechy niekategorialne (takie jak ϕ) mogą dostarczać etykiet w ramach algorytmu etykietowania (*Labeling Algorithm*), co czyni głowę przymiotnikową zbędną. Imiesłowy przymiotnikowe czynne są zatem definiowane jako konstrukcje typu „centaur”, wykazujące właściwości wielu kategorii, co jednak nie wiąże się ze zmianą kategorii syntaktycznej.

Słowa kluczowe

algorytm etykietowania, cechy ϕ , imiesłów przymiotnikowy czynny, język polski, minimalizm, rekategoryzacja, składnia, uzgodnienie.

1. Introduction

This paper investigates the syntax of present active participles in Polish, with particular focus on the contrast between agreeing and invariant forms in *-qć*. Polish present active participles present a striking paradox: despite being morphologically very close and transparently derived from the same verbal stem, they give rise to a clear and systematic syntactic and interpretive split. On the one hand, Polish exhibits invariant participles, such as *czyta-jqć* ‘reading’, which lack agreement morphology, behave as adverbial converbs and are typically interpreted as subject-oriented (as in (1)). On the other hand, Polish also has agreeing participles, such as *czyta-jqć-y* ‘reading’, which display full ϕ -agreement in gender, number and case, are often object-oriented and occur in environments traditionally associated with adjectives (as in (2)).

- (1) Zasta-ł-em Ann-ę, **czyta-jqć** w ogrodzie.
 find-PST-1SG Anna-ACC read-PTCP in garden
 ‘I found Anna (while I was) reading in the garden.’
- (2) Zasta-ł-em Ann-ę, **czyta-jqć-ą** w ogrodzie.
 find-PST-1SG Anna-ACC read-PTCP-ACC.SGF in garden
 ‘I found Anna (who was) reading in the garden.’

This contrast is particularly transparent in perception verb constructions, where invariant and agreeing participles can alternate within the same syntactic frame while yielding sharply different interpretive outcomes. In these

contexts, the agreeing participle (2) typically predicates over the internal argument of the matrix verb, whereas the invariant participle (1) yields a subject-oriented interpretation. Because the two forms share the same verbal stem, aspectual interpretation and local syntactic environment, perception verb constructions provide a controlled and revealing testing ground for identifying the structural source of the distinction, while simultaneously highlighting the substantial overlap between the two participial types.

At first glance, the contrast appears to motivate a categorial division. Invariant participles resemble adverbial converbs, while agreeing participles appear adjectival in both morphology and distribution. This immediately raises the fundamental theoretical question of whether these two participial forms genuinely belong to different syntactic categories, or whether their surface divergence rather obscures a deeper structural unity. Addressing this question has direct consequences for the analysis of Polish participles, but also bears on more general issues concerning syntactic category, recategorization and the division of labor between morphology and narrow syntax.

I argue that Polish present active participles are best analyzed as centaur expressions: hybrid constructions that combine a fully verbal internal structure with morphosyntactic properties typically associated with other categories. More specifically, I propose that both invariant and agreeing participles originate from the same underlying verbal-aspectual structure. The observed contrast does not result from a syntactic recategorization process (from verb to adjective), but instead follows from two independently motivated factors: (i) the structural position in which the participial phrase is merged, and (ii) the presence or absence of an additional ϕ -probe, which licenses agreement without altering the categorial identity of the participle. Under this view, agreeing participles are not adjectival in narrow syntax, despite their adjectival morphology and distribution.

This proposal allows for a unified analysis of the Polish facts while avoiding unnecessary category proliferation. It captures the verbal properties shared by both participial types (including argument structure, aspectual interpretation and sensitivity to verbal diagnostics) while deriving their surface differences from independently motivated functional structure. More generally, the analysis contributes to ongoing debates about whether agreement morphology and adjectival distribution necessarily entail adjectival syntax, or whether such properties can be derived without postulating category-changing operations in narrow syntax.

The broader theoretical claim developed in this paper is that morphology and narrow syntax provide distinct types of evidence. Morphological identity or similarity does not entail categorial identity in syntax, and conversely, syntactic structure may remain verbal even when agreement exponents

are morphologically realized. By adopting a feature-based minimalist architecture in which agreement, Case and selection are mediated by functional heads and probing relations, the analysis captures the Polish participial system while minimizing recategorization and reducing the theoretical burden placed on categorial labels.

The paper is organized as follows. Section 2 introduces the empirical domain, focusing on the distribution and interpretation of invariant and agreeing participles in Polish, with special attention to perception verb constructions. Section 3 develops the theoretical background, addressing the distinction between categorial and non-categorial features, labeling and the cost of recategorization. Section 4 presents a detailed syntactic analysis of Polish participles, deriving their properties from structural position and the presence or absence of an independent ϕ -probe. Section 5 concludes by summarizing the results and discussing their implications for theories of category, agreement and the architecture of grammar.

2. Agreeing and invariant participles in Polish: Core diagnostics

2.1 Agreeing and invariant participles with perception verbs

This section presents a detailed empirical investigation of two types of Polish present active participles, an agreeing participle and an invariant participle. Although the two forms are morphologically closely related and built on the same verbal stem, they differ systematically with respect to agreement, antecedent selection and distribution.¹ At the same time, both retain core verbal properties, such as a progressive aspectual interpretation (particularly in constructions with perception verbs) and the ability to license internal arguments. The goal of this section is to establish the empirical contrast between the two participial forms in Polish, using a uniform set of diagnostics that will later motivate a unified syntactic analysis.

¹ A reviewer pointed out that this distribution finds a typological parallel in the French gerundive and adjectival present participle forms. Specifically, much like the Polish invariant participle in (1), the French gerund is typically subject-oriented and functions as an adverbial modifier of the matrix event. There is, however, a significant point of divergence between Polish and French concerning adjectival forms (2): in French, they only exhibit agreement when they lack verbal complements and function as pure adjectives. When a direct object is present, agreement is usually lost. Polish differs sharply in this regard: as seen in (7) and (8), the participle continues to agree with the head noun even when it licenses internal arguments. This reinforces the ‘centaur’ nature of the Polish form: it preserves its verbal licensing power while simultaneously exhibiting the morphological agreement typically reserved for the adjectival domain.

Perception verb constructions provide a particularly controlled environment for diagnosing the properties of participles. In such contexts, it is possible to observe whether a participle predicates over the internal argument of the matrix verb or instead yields a subject-oriented interpretation. These constructions also make it possible to test whether the presence of agreement morphology correlates with object orientation.

Polish present active participles appear in two distinct forms in perception verb contexts. When the participle bears agreement morphology, it is typically object-oriented and exhibits full agreement in case, number and gender with the internal argument of the matrix verb. This is illustrated in (3).

- (3) Widz-ę Justyn-ę, **wracaj-ąc-a** z koncertu.
 see-1SG Justyna-ACC return-PTCP-ACC.SGF from concert
 'I see Justyna coming back from the concert.' (= *Justyna is coming back from the concert*)

In (3), the participle agrees with the object *Justyna* and predicates over that object. The perceived event is therefore one in which the object participant is the agent of the embedded eventuality. The agreeing participle behaves like a secondary predicate targeting the internal argument of the perception verb.

When the participle appears in its invariant form, by contrast, no agreement morphology is present, and the interpretation shifts accordingly. As shown in (4), the participle now yields a subject-oriented reading.

- (4) Widz-ę Justyn-ę, **wracaj-ąc** z koncertu.
 see-1SG Justyna-ACC return-PTCP from concert
 'I see Justyna while coming back from the concert.' (= *I am coming back from the concert*)

Here, despite the identical linear order and verbal stem, the lack of agreement morphology correlates with a subject-oriented interpretation: the participial clause describes an event associated with the matrix subject rather than the object. Thus, in Polish perception verb contexts, the presence or lack of agreement on the participle correlates directly with its object versus subject orientation.²

² As I will show in Section 4, the shift from object-orientation in (3) to subject-orientation in (4) is not merely a morphological choice but reflects a difference in the syntactic attachment. While the agreeing participle in (3) attaches a secondary predicate to the internal argument, the invariant participle in (4) attaches higher as a clausal adjunct, where it enters into a control relation with the matrix subject.

2.2 Invariant participles as adverbial converbs

Beyond perception verb constructions, the invariant participle in Polish more generally functions as an adverbial converb (an adjunct) expressing an event simultaneous with the event denoted by the main verb (Bojałkowska, 2010; Cetnarowska, 2007). The invariant participle is formed exclusively from imperfective stems and typically corresponds to English *-ing* constructions. This use is illustrated in (5).

- (5) Piotr pi-ł herbat-ę, **czyta-jąc** książk-ę.
 Piotr drink-PST.SGM tea-ACC read-PTCP book-ACC
 ‘Piotr drank tea, reading a book.’

The invariant participle can be replaced by a finite subordinate clause introduced by *gdy* ‘when’, without a change in meaning, as shown in (6).

- (6) Piotr pi-ł herbat-ę, gdy czyta-ł książkę.
 Piotr drink-PST.SGM tea-ACC when read-PST.SGM book-ACC
 ‘Piotr drank tea, when he was reading a book.’

Despite its adverbial behavior, the invariant participle retains verbal properties: it licenses direct objects and prepositional complements and contributes a progressive aspectual interpretation. Crucially, however, it does not exhibit agreement morphology and does not predicate over a nominal argument in the matrix clause.

2.3 Agreeing participles as nominal modifiers

The agreeing participle contrasts with the invariant form in the realization of agreement morphology and in its syntactic distribution, while preserving verbal properties. In Polish, the agreeing participle is morphologically identical to the invariant form apart from the presence of an agreement ending. It can occur either in attributive position, preceding the noun, or in depictive position, following it, as illustrated in (7) and (8).³

- (7) Rozmawia-li-śmy z **pomaga-jąc-y**mi dzieci-om nauczyciel-ami.
 talk-PST-1PL with help-PTCP-INS.MPL child-PL.DAT teacher-INS.PL
 ‘We were talking with teachers helping children.’

³ A reviewer correctly pointed out that the Polish pattern in (7) and (8) is typologically noteworthy when contrasted with Germanic languages like German and Dutch. In those languages, agreement on the present participle is typically restricted to the prenominal position. In postnominal contexts, the participle remains invariant. Polish, by contrast, requires full ϕ -agreement regardless of the participle’s linear position relative to the noun. As I will further develop in Section 4, this suggests that in Polish the licensing of a ϕ -probe is tied to the internal structure of the participial phrase itself rather than its specific landing site.

- (8) Rozmawia-li-śmy z nauczyciel-ami **pomaga-jąc-ymi** dzieci-om.
 talk-PST-1PL with teacher-INS.PL help-PTCP-INS.MPL child-PL.DAT
 ‘We were talking with teachers helping children.’

In both configurations, the participle predicates over (and agrees with) the noun it modifies, which corresponds to the internal argument of the matrix verb. At the same time, it licenses verbal complements and contributes a progressive or continuous aspectual meaning. These properties indicate that, despite its adjectival distribution and agreement morphology, the agreeing participle preserves a fundamentally verbal structure.

In sum, the Polish data establish a clear empirical contrast between invariant and agreeing present active participles. The two forms differ systematically in agreement and syntactic orientation, yet share core verbal properties. This pattern raises the central analytical question addressed in the remainder of the paper: how agreement and nominal modification can arise without invoking a change in syntactic category from verb to adjective.

2.4 Additional diachronic and comparative evidence

The empirical pattern observed with Polish present active participles can be further strengthened by considering a broader range of internal diagnostics. Taken together, these diagnostics converge on the conclusion, discussed in Sections 3 and 4, that apparent adjectival distribution and agreement morphology do not require syntactic recategorization of a verbal structure. Instead, Polish provides independent evidence that surface distribution, agreement and internal syntax may diverge and that participial forms can retain a verbal core even when they appear in positions canonically associated with adjectives.

The first source of evidence comes from the historical distribution of invariant participles. In Old Polish, invariant present active participles were permitted in positions that are synchronically restricted to agreeing adjectival modifiers, including both pre-nominal and post-nominal environments. Crucially, these participles did not exhibit adjectival agreement morphology, even when they occurred adjacent to fully inflected adjectives. This is illustrated by Klemensiewicz et al. (1985) in the 15th-century examples in (9)–(10).

- (9) ...żem widzia-ł **umiera-jąc** mił-e dziecię moj-e...
 that-1SG see-PST.SGM dying-PTCP dear-NSG child my-NSG
 ‘...that I have seen my dear dying child..’

(Kazania gnieźnieńskie, 15th c.)

- (10) Nalaz-ł dziewic-ę Mari-ę **siedz-ąc...**
 find-PST.SGM virgin-ACC Mary-ACC sit-PTCP
 ‘He found the virgin Mary sitting..’

(Rozmyślanie przemyskie, 15th c.)

In (9), the invariant participle *umierając* appears before the fully agreeing adjective *mile*, without itself showing any agreement morphology. These data demonstrate that invariant participles could historically occupy attributive and depictive positions that would otherwise be analyzed as adjectival, without exhibiting any morphological trace of recategorization. This strongly suggests that adjectival distribution alone is insufficient evidence for an adjectival syntactic category.⁴

Further support for a verbal analysis comes from adverbial modification. Polish sharply distinguishes adverbs that modify verbal phrases from those that target adjectival structures. Verbal predicates and participles are compatible with modifiers such as *dużo* ‘much/a lot’, whereas adjectives require degree modifiers such as *bardzo* ‘very’. This contrast is illustrated in (11)–(12).

- (11) Dużo mówiący-y polityk.
 much speak-PTCP-NOM.SGM politician
 ‘A politician speaking a lot.’

- (12) Bardzo gadatliw-y polityk.
 very talkative-NOM.SGM politician
 ‘A very talkative politician.’

The acceptability of *dużo* in (11) indicates that the participle represents a verbal structure capable of hosting VP-modifying adverbs. By contrast, the incompatibility of *bardzo* with participles such as *mówiący* shows that no adjectivizer is present. This contrast provides a clear syntactic diagnostic distinguishing verbal participles from lexical adjectives, reinforcing the claim that agreeing participles do not merge an adjectival layer.⁵

Additional evidence comes from the contrast that can be drawn with South Slavic languages such as Serbian, where formally similar participial forms have undergone full lexicalization as adjectives.⁶ This difference is illustrated in (13)–(15).

⁴ The invariant behavior of Old Polish participles in (9) and (10) provides a link to the synchronic state of languages like German and Dutch, where postnominal participles similarly resist agreement. This suggests that the mandatory agreement observed in Modern Polish is a relatively recent development in the history of the centaur construction.

⁵ A similar pattern is observed again in German and Dutch; despite bearing adjectival agreement morphology in attributive positions, participles in these languages are modified by verbal adverbs rather than adjectival degree modifiers. This cross-linguistic evidence supports the claim that agreement morphology is not a definitive diagnostic for a shift to the adjectival category.

⁶ Notice that Serbian expresses structures with perception verbs parallel to the Polish ones discussed in this paper by means of finite complement clauses (Frasson, 2025).

- (13) Szybko jad-ąc-y po mieście samochód.
 fast go-PTCP-SGM in town car
 ‘A car driving fast around the town.’

(Polish)

- (14) *Brzo id-uć-i po gradu auto.
 fast go-PTCP-SGM in town car
 Intended: ‘A car driving fast around the town.’

(Serbian)

- (15) Idući autobus.
 next bus
 ‘The next bus.’

(Serbian)

In Polish, the participle in (13) retains full verbal properties, including argument structure and manner modification. In Serbian, by contrast, comparable forms have lexicalized as adjectives, as shown by their restricted semantics and lack of verbal behavior. This contrast highlights the central claim of this paper: participial morphology does not uniquely determine syntactic category. Rather, languages may differ in whether a given participial form retains a verbal core or undergoes true adjectival recategorization.⁷ The central question is therefore why Polish has maintained a centaur system where agreement does not trigger a loss of verbal properties, unlike the recategorization observed in the Serbian system described above.⁸

In sum, the Polish data show that agreement morphology, adjectival distribution and (historically) placement do not constitute sufficient evidence for syntactic recategorization. Instead, a feature-driven analysis in which participles retain a verbal-aspectual core, while agreement features are independently licensed, provides a more economical and empirically adequate account.

⁷ A reviewer pointed out that a similar, even more formalized split is found in French, which distinguishes between the verbal present participle and the verbal adjective. This split is sometimes marked orthographically (e.g., *précédant* ‘preceding’, verb; *précédent* ‘preceding’, adjective) and is rooted in a distinct diachronic development where two separate Latin forms (the ablative gerund in and the imperfective participle) merged morphologically but remained syntactically distinct (see Živojinović, 2021 for an overview).

⁸ It should be noted that certain participles, such as *krytykujący* (‘criticizing’/‘critical’) or *przesadzający* (‘exaggerating’), may occasionally appear with the degree modifier *bardzo* in specific contexts. However, this occurs only when the participle undergoes lexicalization, shifting its meaning from an ongoing action to a stable quality. Crucially, in such cases, the participle typically loses its ability to license a direct object. This supports the claim that the productive participial spine is verbal; the appearance of *bardzo* signals a loss of that verbal structure and a shift into the class of true lexical adjectives, similar to the Serbian cases discussed in (15).

2.5 Research questions and goals

In view of the data discussed in Section 2, the first empirical question to be addressed in this paper concerns the source of the contrasting interpretation between the two types of participle (invariant and agreeing) despite being built on the same verbal stems and sharing nearly identical morphological material.

A closely related question concerns the grammatical status of this distinction. In particular, the paper asks whether the contrast between invariant and agreeing participles reflects a genuine difference in syntactic category, corresponding to an adverbial versus adjectival opposition, or whether it should instead be understood as the result of different structural configurations built from the same underlying verbal syntax. Put differently, the question is whether the observed surface differences require distinct categorial heads or recategorization operations in the narrow syntax, or whether they can be derived without positing a change in category.

Beyond the empirical contrast itself, the paper addresses a set of syntactic questions concerning the internal composition and derivation of participial structures. A central issue is the internal structure of Polish present active participles and, more generally, the size and content of the syntactic structure that underlies both invariant and agreeing forms. Relatedly, the paper asks where in the syntactic derivation the two participle types are merged and whether they occupy distinct structural positions or are introduced at different stages of the clausal spine. Finally, the paper investigates the role played by ϕ -agreement in shaping both the distribution and the interpretation of participles, with particular attention to how agreement correlates with orientation and predication in perception verb contexts.

The primary goal of the analysis is to develop a feature-based minimalist account of Polish present active participles that captures the empirical contrast between invariant and agreeing forms without multiplying syntactic categories or derivational mechanisms. The analysis aims to show that invariant and agreeing participles share the same underlying verbal syntax, including their aspectual and argument-structural properties, and that the difference between them arises from independently motivated factors. In particular, the account attributes the contrast to the presence or absence of an independent ϕ -probe and to the structural position in which the participial structure is merged, rather than to any difference in categorial status.

A further goal is to demonstrate that no V-to-A recategorization is required in narrow syntax to derive the adjectival properties of agreeing participles. Instead, adjectival behavior is shown to emerge from the interaction between verbal structure, agreement and structural embedding. In this respect, the analysis seeks to dissociate agreement morphology from categorial

identity, arguing that the presence of ϕ -agreement does not in itself entail a change from verbal to adjectival category.

More broadly, the paper aims to clarify the theoretical distinction between categorial and non-categorial features in the syntax. Categorial features, such as V, N and A, are taken to be relevant primarily for External Merge and the basic combinatorial possibilities of lexical items. Non-categorial features, including ϕ -features, aspectual features and Case, are instead argued to drive syntactic operations such as Agree and to shape the distribution and interpretation of syntactic objects without determining their categorial identity. By maintaining a strict separation between these two types of features, the analysis seeks to reduce the explanatory burden placed on categorial labels in the syntax.

Finally, the paper connects the proposed syntactic account to broader considerations concerning the processing cost of recategorization. To the extent that recategorization operations impose additional computational or processing demands, the analysis argues that their role in syntactic theory should be minimized whenever possible. The feature-based approach developed here thus contributes to a view of syntactic structure in which categorial labels play a limited role, and where surface differences between constructions are derived from the interaction of features and structure rather than from category-changing operations.

3. Features and categories

3.1 Towards a minimal(ist) structure of active participles

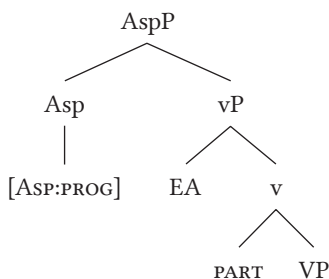
This section outlines the structural assumptions that will serve as the point of departure for the analysis developed in the remainder of the paper. I begin by adopting a minimal(ist) syntactic representation of active participles that is compatible with both their verbal properties and their morphosyntactic realization, while remaining agnostic as to whether participial structures must be categorially adjectival in the narrow syntax. Specifically, I adopt the architectural assumptions of Halle and Marantz (1994). Under this view, narrow syntax manipulates abstract feature bundles rather than fully formed words. The adjectival appearance of agreeing participles is a result of a late insertion principle, where morphological exponents are inserted into functional heads at the interface.

I assume that active participles are built on a verbal spine in which the lexical verb corresponds to a VP that merges with a vP. The vP introduces the external argument of the predicate in its specifier, in line with Embick (2004) and Collins (2005). In addition to its role in argument structure, v is

assumed to carry a *PART* feature, following D'Alessandro and Roberts (2010). This feature is responsible for the participial morphology observed on the verbal stem. Importantly, the presence of *PART* is not taken to imply any categorical shift; rather, it is a feature that receives morphological exponence at Spell-Out. Above *vP*, the structure includes an aspectual phrase, *AspP*, which hosts a progressive aspectual feature. This captures the well-known aspectual properties of present active participles, which systematically correspond to progressive or ongoing eventualities.

Under these assumptions, the minimal structure of an active participle can be represented as in (16). The participial phrase consists of an *AspP* dominating a *vP*, which in turn dominates *VP*. The external argument is merged in *Spec,vP*, while participial morphology is associated with the *PART* feature on *v*, and progressive aspect is encoded in *Asp*.

(16)



Structures of this kind have traditionally been analyzed for gerunds and participles as ‘reduced’ relative clauses, particularly in the literature on participial modification and agreement (see for instance Belletti, 1992). While the present proposal does not deny the descriptive adequacy of this tradition, it departs from it in refraining from encoding reduction status directly in the syntactic representation.

A prominent alternative view in the literature holds that participial phrases become adjectival by merging with a dedicated adjectival head. On this morphosyntactic approach, developed in various forms by Alexiadou et al. (2014), Borik and Gehrke (2019), Embick (2004), Gehrke (2015), Marín and Fábregas (2021) and Sleeman (2011), participles are verbal or aspectual phrases that are converted into adjectival phrases when selected by an adjectivizing head, typically labeled *a*. This head is assumed to take a *vP* or *AspP* as its complement and to merge an *aP*, thereby endowing the participial structure with adjectival properties.

Within this line of work, participles of different structural sizes are claimed to be eligible for adjectivization, and the presence of adjectival morphology and distributional properties is taken as evidence for the syntactic presence

of an aP layer. On some accounts, this reasoning is extended to all active participles, including English *-ing* forms, which are argued to be adjectival across the board (as proposed, among others, in Bešlin (2026)). Empirical arguments in favor of this position include the observation that participial agreement suffixes often match adjectival morphology, and that participles can appear in contexts otherwise reserved for adjectives. In addition, it is sometimes claimed that adjectivized participles no longer license aspectual structure.

While the present analysis does not deny the empirical relevance of categorial heads in the grammar, several considerations argue against making an aP layer obligatory in the case at hand. First, within a minimalist framework, morphology must be kept distinct from syntax. Agreement morphology, in particular, may reflect post-syntactic realization rather than syntactic category, and morphological similarity alone cannot be taken as decisive evidence for categorial identity. As recently argued independently by Talsma (2025), surface morphology does not uniquely determine the syntactic structure of a participle.

Second, in the specific case of Polish active participles, a recategorization analysis risks substantial overgeneration. The participles under discussion retain clear aspectual interpretations, most notably progressive aspect, which is difficult to reconcile with the claim that aspect is systematically unavailable once adjectivization has applied. Moreover, the morphological evidence for adjectival status is inconclusive. Crucially, participles preserve verbal argument structure, as illustrated by examples such as (7) and (8), where the dative complement is licensed in a way that is fully consistent with a verbal syntax and difficult to explain if we assume that participles are really adjectival.

Finally, the presence of ϕ -agreement does not in itself entail a shift in syntactic category. Agreement can be independently licensed by a ϕ -probe hosted on a functional head, without requiring the introduction of an adjectivizer. If ϕ -agreement can be derived through standard mechanisms of Agree, then there is no principled reason to postulate an aP layer solely to account for agreement morphology. This observation opens the possibility that agreeing participles remain verbal in the narrow syntax, differing from invariant participles not in category, but in their structural position and the availability of agreement.

The conclusion that agreement and distributional properties do not entail a change in syntactic category is consistent with approaches that minimize or eliminate lexical categories from narrow syntax altogether. In particular, De Belder (2011) argues that categories such as adjectives are not syntactic primitives, but emerge from independently motivated functional structure. On this view, morphosyntactic properties traditionally taken to

diagnose category membership, such as agreement morphology or adjectival distribution, cannot by themselves justify recategorization. This perspective provides a natural theoretical backdrop for the present analysis, in which Polish active participles retain a verbal core while exhibiting adjectival surface behavior as a result of feature-driven mechanisms rather than category-changing operations.

These considerations motivate the approach pursued in the remainder of the paper, in which active participles are uniformly built as verbal-aspectual structures, and apparent adjectival behavior is derived without recourse to obligatory adjectivization in the syntax.

3.2 Labels, categories and the cost of recategorization

A central theoretical assumption underlying the present analysis concerns the distinction between labels and syntactic categories, and the role of labeling in making syntactic objects interpretable at the interfaces. Following Chomsky (2013), syntactic objects must be assigned a label in order to be legible at Logical Form (LF). Crucially, however, the label required for interpretability need not correspond to a traditional lexical category such as V, N or A. What LF requires is a feature that allows interpretation, not categorial identity *per se*.

Within this framework, there is a Labeling Algorithm (LA) that computes labels on the basis of prominent features available within a syntactic object. These may include categorial features, but they may equally consist of non-categorial features such as φ -features, aspectual features or functional heads like *v*. Once the LA identifies an appropriate feature to serve as the label, the syntactic object becomes interpretable at LF. Importantly, the assignment of such a label does not in itself entail recategorization. Labeling and categorization are therefore distinct: labeling is an interface requirement, whereas syntactic category need not be encoded directly in narrow syntax.

This distinction has immediate consequences for the analysis of participial constructions. In particular, additionally merging a φ -head in the participial structure in (16) is sufficient to provide a prominent feature, functioning as a label that renders the structure legible at LF. The presence of φ -features allows agreement relations to be established and interpreted, while the underlying syntactic structure may remain fully verbal.

Under this view, there is no principled necessity to posit an adjectival phrase (aP) merely in order to license agreement or interpretation. A non-categorial label supplied by φ suffices.

Independent support for this conclusion comes from considerations of computational and processing cost. In minimalist syntax, Merge is assumed to be a cost-free operation used to build structure (Chomsky, 2013).

By contrast, changing the syntactic category of an object involves operations that go beyond narrow syntax. As shown in Partee (1987), type-shifting and recategorization have been argued to involve semantic reinterpretation and increased complexity at the interfaces. Such operations are therefore not expected to be cognitively or computationally free at the narrow syntactic level.⁹

Psycholinguistic evidence provides converging support for this asymmetry. A substantial body of research on sentence processing shows that temporary misclassification of syntactic category leads to increased processing difficulty, often manifested as garden-path effects (Frazier & Rayner, 1982; Pritchett, 1988). When a form is ambiguous between a verbal and an adjectival or nominal interpretation, comprehenders tend to initially parse it as verbal. If later material forces a non-verbal interpretation, the parser must revise the structure, incurring a measurable processing cost.

This effect is well documented for English participial forms. For instance, in expressions such as *the cat meowing in the garden*, the participle *meowing* is initially compatible with a verbal parse. Only in configurations such as *the meowing cat in the garden* does the adjectival interpretation become unambiguous from the outset. Experimental studies show that in ambiguous cases, readers overwhelmingly adopt a verbal analysis first, and experience difficulty when subsequent context forces an adjectival or nominal reading (Ferreira et al., 2002; Hale, 2001; Trueswell et al., 1993). These garden-path effects indicate that recategorization is not cognitively neutral: it requires reanalysis and imposes additional load during real-time parsing.

Taken together, these considerations argue against positing recategorization unless it is independently motivated. If a structure can be interpreted without changing its underlying category, by means of functional heads and feature-driven labeling, such an analysis is both theoretically and cognitively preferable. Recategorization appears to involve greater cognitive and structural effort than the mere extension of functional structure, and may instead be associated with processes of lexicalization, such as the loss of internal verbal structure and the reanalysis of upper layers as adjectival.

This perspective directly motivates the analysis developed in Section 4. In the absence of clear evidence that Polish agreeing participles do not exhibit real verbal syntax (given their preservation of argument structure, aspectual

⁹ While Partee (1987) focuses on semantic type-shifting within the nominal domain, I invoke this reference here as a broader theoretical parallel. The core idea is that operations that alter the interpretative status of a constituent (whether through semantic type-shifting or syntactic recategorization) are generally assumed to be more complex than those that proceed via standard Merge and Agree. In this sense, the centaur analysis is more economical because it avoids the additional mapping requirements that a V-to-A shift would impose at the syntax-semantics interface.

interpretation and verbal diagnostics) there is no principled reason to assume a change in syntactic category. Differences in agreement and distribution can instead be derived through variation in the presence of functional heads, such as an additional φ -probe, without invoking adjectival recategorization. More generally, the cost associated with recategorization provides a strong theoretical reason to favor analyses that minimize category change whenever narrow syntactic tools suffice.

4. Analysis

4.1 Participle structures and the extra φ -probe

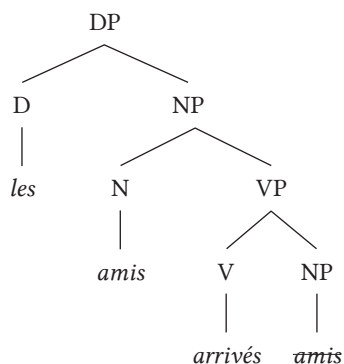
In the analysis of Polish present active participles, I build on previous analyses of French reduced participial clauses (Belletti, 1992; Cecchetto & Donati, 2020), such as in (17), which provide a valuable point of comparison for understanding the syntactic behavior of the Polish data:

- (17) Les amis arriv-és hier.
 DEF friends arrive-PTCP yesterday
 ‘The friends (that) arrived yesterday.’

As observed by Belletti (1992) and further discussed in Cecchetto and Donati (2020), these forms are not adjectival participles; rather, they are reduced relative clauses that maximally realize a VP. The apparent adjectival behavior in surface syntax arises from NP movement out of the VP, analogous to a reduced relative clause. The movement of the NP to a higher position within the DP accounts for the surface ‘relabeling’ effect: although the participle appears in what could be interpreted as a postnominal adjectival position, the underlying structure is fully verbal. This analysis captures the empirical pattern that participles retain verbal syntax even when superficially appearing to function adjectivally.¹⁰

¹⁰ While I follow Belletti (1992) in treating these as reduced structures, I emphasize that the relabeling in Polish is achieved via φ -features at the interface, rather than a category-changing head. This captures the fact that, unlike French agreeing participles which often behave as pure adjectives, Polish participles remain ‘centaurs’ that host both agreement and active verbal licensing.

(18)



Extending this reasoning to Polish, I propose a feature-driven syntactic computation in which the syntactic structure of both agreeing and invariant participles is initially identical. However, the analysis proposed here departs from the one argued for the structure in (18) in one fundamental way. Under a minimalist approach in which Merge and Agree drive syntax and surface morphology is determined post-syntactically (Borer, 2005), the two participle types differ solely with respect to the presence or absence of an additional φ -probe, and no reduced relative clause structure needs to be assumed.

The φ -probe is an independent functional head that searches its c-command domain for a nominal goal. Critically, it does not introduce a new argument or assign Case; its sole function is to trigger agreement through its unvalued φ -features, thereby producing the morphologically agreeing form (see Coon & Preminger, 2012; D'Alessandro, 2017; Halpert, 2015; van der Wal, 2022 for similar approaches). In the absence of this probe, the participle remains invariant, exhibiting purely verbal morphology and aspectual interpretation.

In morphosyntactic terms, the additional φ -probe behaves similarly to an adjectivizer: it enables the participial phrase to receive agreement morphology that is compatible with adjectival paradigms, without invoking a syntactic recategorization of the verbal structure. Consequently, the structural and interpretive differences between agreeing and invariant participles can be captured through independent feature-driven mechanisms rather than through a shift in categorial status. This analysis preserves the verbal core of the participle, accounts for the distributional patterns observed across multiple languages and aligns with the minimalist tenet that narrow syntax operates with the simplest possible operations, while surface morphology and categorial interpretation are determined at the interface.

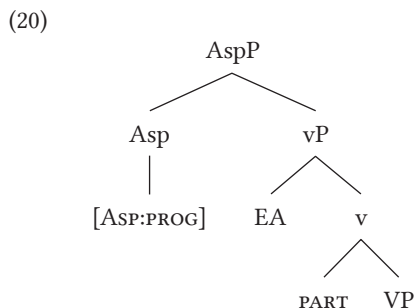
4.2 The post-NP agreeing form

I illustrate the derivation of post-NP agreeing participles with the Polish example in (19), where the participial phrase appears after the NP and exhibits full agreement in gender, number and case:

- (19) Dziewczyna **śpiew-ając-a** piosenkę.
 Girl sing-PTCP-SGF song
 ‘The girl singing a song.’

As discussed in Section 3, the participial constituent in (19) forms a small clause, consisting of a nominal argument and a verbal predicate. The derivation begins with the merge of the verbal core. First, VP merges with vP. The external argument is introduced in Spec,vP, following standard assumptions about argument introduction. At this stage, the structure is fully verbal: the participial predicate represents a verbal spine and assigns thematic roles in a manner identical to finite verbal predicates.

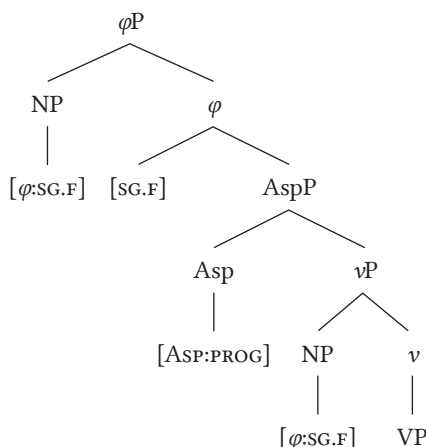
The next step in the derivation involves the introduction of aspectual structure. An AspP is merged on top of vP, carrying a progressive aspectual feature. This phrase encodes the ongoing interpretation characteristic of present active participles and is responsible for anchoring the event structure of the predicate. Importantly, the addition of AspP does not alter the categorial status of the participial phrase: the structure remains verbal-aspectual, and the external argument remains in Spec,vP. The structure is repeated in (20):



At this point, an independent functional head hosting ϕ -features is merged above AspP. This ϕ -head bears unvalued ϕ -features and probes its c-command domain for a suitable goal. The NP introduced in Spec-vP falls within this domain and values the ϕ -features on the probe via Agree, moving to the specifier of ϕ P, possibly in connection with the presence of an EPP-like feature. The ϕ P is merged as the highest functional layer of the participial spine. Its merger is directly triggered by the need to value the ϕ -features of the internal argument. At Spell-Out, the morphological component sees a head with valued ϕ -features and inserts an agreement suffix from the adjectival

paradigm. Crucially, this happens without the need for a category-changing adjectivizer, maintaining the modular separation between syntactic category and morphological exponence. As a result, the participial predicate is morphologically realized with gender, number and case agreement corresponding to the NP.

(21)



The movement of the NP to Spec- ϕ P is triggered by an EPP-feature on the ϕ -head. This movement is not merely mechanical; it serves an interface requirement by ensuring that the ϕ -features are prominent enough to provide a label for the constituent under the Labeling Algorithm. Furthermore, this structural configuration correlates with a predicative interpretation, distinguishing it from the prenominal position, without necessarily leading to re-categorization. Crucially, the introduction of ϕ -features does not involve the merge of an adjectival category. The ϕ -head is a purely functional head whose role is to license agreement, not to recategorize the verbal structure. Once ϕ -agreement has been established, the NP raises to Spec- ϕ P. This movement satisfies interface requirements associated with labeling and visibility, and it yields the surface post-NP configuration.

The resulting structure is therefore labeled as ϕ P at the interface. Following the Labeling Algorithm, the presence of an overt ϕ -head suffices to render the syntactic object interpretable at LF. Aspectual features remain accessible within the structure, but they do not determine the label of the constituent. In this sense, agreement, rather than aspect, plays the decisive role in establishing the interpretability of the participial small clause, even though aspect remains an integral part of its internal structure.

At this point, the reader may wonder how case assignment is mediated by the matrix verb or preposition, as in the cases of (22) and (23). In these

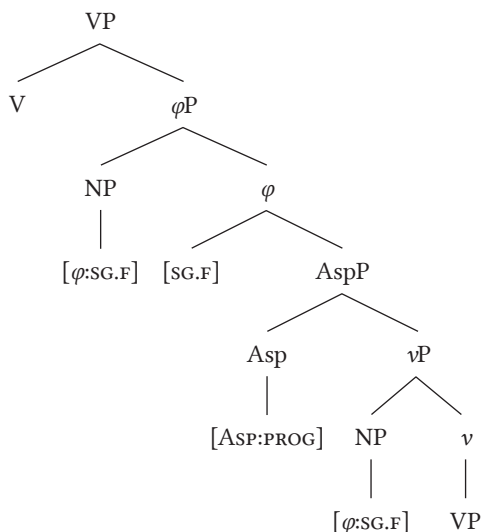
examples, the participle itself does not assign Case; rather, the matrix predicate selects the small clause and assigns Case to the ϕ P, which licenses the participle's case morphology too:

- (22) Słyszy-my dziewczyn-ę **śpiewa-jąc-ą** piosenkę.
 hear-1PL girl-ACC sing-PTCP-ACC song
 'We hear the girl singing a song.'

- (23) Z dziewczyn-ą **śpiewa-jąc-ą** piosenkę.
 with girl-INSTR sing-PTCP-INSTR song-ACC.SG.F
 'With the girl singing a song.'

In (22), the matrix verb c-selects the small clause, which contains the NP. The NP in Spec, ϕ P is accessible to Agree for both Case and ϕ -feature valuation, allowing the participle to surface with full agreement. Notice that the same type of reasoning captures case assignment to the small clause by a preposition (23).

(24)



This derivation raises the question of category selection and how a verb or preposition can select a participle if there is no adjectival or nominal layer. Following the insights of Svenonius (1994), selection operates not on syntactic categories but on features encoded on heads. A selector (V or P) probes its complement for a functional head that carries the necessary ϕ -features and/or Case features. In the present structure, the ϕ P contains valued ϕ -features, satisfying the selection requirement of the higher predicate without invoking an adjectival or nominal layer. Selection is thus mediated via feature matching, not categorial identity.

Long-distance Agree formalizes this relation: a selector *X* bears an unvalued feature [*uF*] and probes its *c*-command domain for a head *Y* with matching [*F*]. Once matched, the selection is satisfied. Critically, the selector does not probe for an *aP* or *nP*; it targets the functional head where the relevant features reside. Accordingly, the derivation requires no recategorization at the narrow syntactic level, as all necessary agreement and selection mechanisms operate through feature-driven Agree.

Finally, Case assignment is fully integrated into this mechanism. As *C*-selection can target features such as Case (Svenonius, 1994), the matrix verb or preposition assigns Case to the NP in Spec- ϕ P, which is simultaneously the locus of ϕ -feature valuation for the ϕ -probe. This dual mechanism licenses the participle's morphology and ensures compatibility with selection by higher predicates. Importantly, this analysis demonstrates that post-NP agreeing participles can be fully accounted for by the present approach, without invoking overt adjectival or nominal layers, preserving both the verbal core and the aspectual interpretation of the participle.

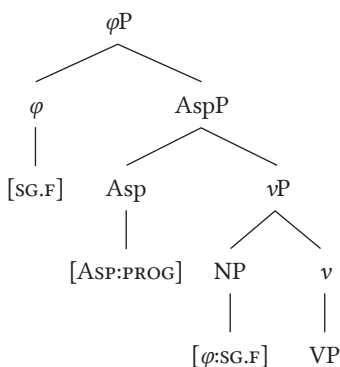
4.3 The pre-NP agreeing form

Pre-NP agreeing participles, as illustrated in (25), provide a surface variant in which the participial phrase precedes the NP, yet still displays full agreement in gender, number and case:

- (25) **Śpiewa-jąc-a** piosenkę dziewczyna.
 sing-PTCP-SGF song girl
 ‘The girl singing a song.’

The derivation is largely parallel to the post-NP form: the internal argument NP is merged in Spec-*vP* within a small clause, and an independent ϕ -probe *c*-commands the *vP*, licensing agreement through Agree. The syntactic computation involves again movement of the verb to *v* and then to *Asp*, to pick up the participial suffix and value the aspectual features. This movement ensures that the verbal head precedes its internal arguments within the participial phrase, as shown in (25). In addition, I propose that, unlike the post-NP configuration, the NP does not move to Spec- ϕ P. This captures the fact that the participle surfaces before the NP without requiring an overt adjectival or nominal layer. The structure relies on the optionality of EPP-like features, consistent with the proposals in Cable (2012), Fanselow (2004) and Müller (2001), where the EPP is formally distinct from ϕ -features and agreement need not trigger movement. As a result, ϕ -Agree can apply independently, allowing the NP to remain in situ while the participle receives fully valued agreement morphology.

(26)



In this configuration, the participle occupies a surface pre-NP position due to the relative ordering of the functional heads and does not require the NP to move to the specifier of φ P. The φ -probe still ensures that the participle exhibits correct gender, number and case agreement, while selection by a matrix verb or preposition targets the functional head containing these features rather than an adjectival or nominal category. The Case of the NP is assigned by the higher predicate as in the post-NP scenario, ensuring morphosyntactic well-formedness.

This analysis demonstrates that the surface pre-NP position of the participle can be derived without any additional movement of the NP, relying solely on the feature-driven Agree mechanism. It provides a unified account of both pre- and post-NP agreeing participles, showing that their surface distribution is compatible with a purely verbal structure extended by φ P, without invoking categorial recategorization. This approach preserves the verbal syntax and aspectual interpretation while systematically explaining agreement, Case assignment and selection.

4.4 The invariant form

Invariant participles, illustrated in (27), differ fundamentally from agreeing participles in that they do not carry a φ -probe and do not license a surface subject NP in vP.

- (27) Czyta-jąc książk-ę.
 read-PTCP book
 ‘Reading a book.’

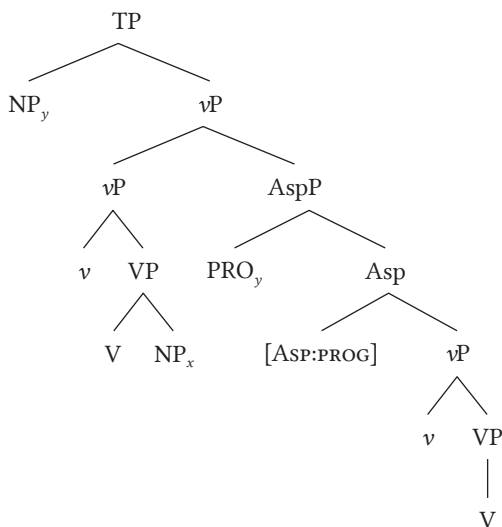
In this construction, the aspectual small clause functions as an adjunct to vP rather than a core argument-bearing small clause. The subject of the participle is a controlled PRO, merged in Spec,AspP,¹¹ which receives its reference

¹¹ In line with Bošković (2012), I assume that Spec,AspP can function as a subject position when Spec,TP is not realized.

from an external controller (often the matrix subject). Because there is no ϕ -probe c-commanding the vP, agreement cannot be established, and because an NP is missing, the structure lacks Case features. Consequently, the participle surfaces as an invariant form, with no morphological agreement in gender, number or case.

The syntactic structure can be represented as follows:

(28)



Here, the PRO in Spec,AspP is controlled by the matrix subject NP, ensuring interpretation without triggering agreement. The adjunct status of the invariant small clause guarantees that the participle does not participate in Case or ϕ -valuation with the matrix predicate. As a result, invariant participles display adverbial properties, marking a progressive event simultaneous with the main verb, but without adjectival agreement or structural recategorization.¹² The question remains as to why the ϕ -probe is absent in these cases. In the invariant construction, the participle functions as an adjunct rather than a selected small clause. Because its subject is PRO, it does not require Case or ϕ -valuation from an external probe and the grammar avoids the merge of an unnecessary ϕ -head. This follows a general principle of structural economy:

¹² As correctly pointed out by a reviewer, the availability of this control relation is subject to specific thematic constraints. As noted by Landau (2010) and Żychliński (2011), Experience-type arguments in Polish do not easily control into participial adjuncts. This suggests that the subject-oriented nature of invariant participles is restricted to high-attaching adjuncts that target an external argument with specific thematic properties, further reinforcing the idea that structural position and configuration are the primary drivers of participial interpretation.

functional structure is only merged when required for licensing or labeling overt arguments.

This analysis supports a feature-driven, minimalist syntactic computation, in which invariant and agreeing participles share a common verbal core but differ solely in the presence or absence of the additional φ -probe. It also provides a principled account of why invariant participles can be freely replaced by subordinate clauses (e.g., introduced by *gdy* ‘while’) without affecting interpretation, since their surface position is adjunctive and not argument-bearing.

5. Conclusion

This paper has examined the syntax of Polish participles, with particular attention to the contrast between present active agreeing and invariant forms in *-qc*. Although these participles are morphologically closely related and transparently derived from the same verbal stem, they exhibit a systematic syntactic and interpretive split. Agreeing participles display full φ -agreement in gender, number and case and typically surface in pre- and post-NP configurations, while invariant participles lack agreement morphology and function as adverbial converbs. The central empirical finding is that this contrast does not reflect a difference in underlying syntactic category, but rather follows from independently motivated structural and featural properties.

I have argued that both agreeing and invariant participles are derived from the same verbal core and represent the same verbal-aspectual structure. The difference between the two constructions is not the result of syntactic recategorization (e.g., V-to-A), but instead arises from two factors: the presence or absence of an additional φ -probe and the structural position of the participial small clause. Agreeing participles involve an independently merged φ -head, which establishes an Agree relation with a nominal goal and yields overt agreement morphology. Invariant participles, by contrast, lack this additional probe and therefore surface without agreement. Independently, the distinction between agreeing and invariant participles accounts for their divergent distribution and interpretive properties, including object-oriented versus subject-oriented readings in perception verb contexts. At no point does the derivation require the merge of an adjectival functional layer (aP).

The Polish data thus provide strong evidence that adjectival distribution and agreement morphology are not reliable diagnostics for syntactic category. Invariant participles may occur in positions otherwise associated with adjectives, both synchronically and historically, while retaining verbal modification patterns, argument structure and aspectual interpretation. Conversely, agreeing participles may display adjectival agreement without losing their

verbal core. These facts argue against analyses that equate agreement morphology or surface distribution with categorial identity. In this proposal, the Polish active participle is a centaur precisely because its syntactic spine is verbal, but its morphological output is determined by functional features that are usually shared with adjectives.

From a typological perspective, the Polish system highlights a unique stage in the evolution of participles. Unlike South Slavic languages (e.g., Serbian), where similar forms have often undergone full lexicalization and lost their internal verbal syntax to become true adjectives, Polish centaurs maintain a productive verbal core. This suggests that the presence or absence of the φ -probe is a key micro-parametric variable in the Slavic participial system.

In conclusion, the findings provide support in favor of a modular view of grammar in which narrow syntax, morphology and interpretation act at separate levels in the architecture of grammar. Morphological agreement can be licensed independently of categorial status, and surface similarities between participles and adjectives need not correspond to syntactic recategorization. A feature-based, minimally recategorizing approach allows for a unified analysis of Polish present active participles, capturing both their shared verbal properties and their systematic surface variation. More generally, the Polish system illustrates how such hybrid or ‘centaur’ constructions can be modeled without proliferating syntactic categories, by attributing surface diversity to the interaction of functional features and structural position rather than to category-changing operations in narrow syntax.

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References

- Alexiadou, A., Gehrke, B., & Schäfer, F. (2014). The argument structure of adjectival participles revisited. *Lingua*, 149, 118–38. <https://doi.org/10.1016/j.lingua.2014.05.018>
- Belletti, A. (1992). Agreement and case in past participle clauses in Italian. In T. Stowell & E. Wehrli (Eds.), *Syntax and semantics: Syntax and the lexicon* (pp. 21–44). Brill. https://doi.org/10.1163/9789004373181_004
- Bešlin, M. (2026). Active participles are (deverbal) adjectives. *Journal of Linguistics*, 62(2), 261–94. <https://doi.org/10.1017/S0022226725000155>

- Bojałkowska, K. (2010). *Opis składniowy imiesłowów przysłówkowych we współczesnym języku polskim*. Wydawnictwo Naukowe Uniwersytetu Mikołaja Kopernika.
- Borer, H. (2005). *Structuring sense 2: The normal course of events*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199263929.001.0001>
- Borik, O., & Gehrke, B. (2019). Participles: Form, use and meaning. *Glossa: A Journal of General Linguistics*, 4(1), 1–27. <https://doi.org/10.5334/gjgl.1055>
- Bošković, Ž. (2012). On NPs and clauses. In G. Grewendorf & T. E. Zimmermann (Eds.), *Discourse and grammar: From sentence types to lexical categories* (pp. 179–246). De Gruyter Mouton. <https://doi.org/10.1515/9781614511601.179>
- Cable, S. (2012). The optionality of movement and EPP in Dholuo. *Natural Language & Linguistic Theory*, 30(3), 651–97. <https://doi.org/10.1007/s11049-012-9172-6>
- Cecchetto, C., & Donati, C. (2020). Relabeling participial constructions. In L. Franco & P. Lorusso (Eds.), *Linguistic variation: Structure and interpretation* (pp. 149–62). De Gruyter Mouton. <https://doi.org/10.1515/9781501505201-009>
- Cetnarowska, B. (2007). Departicipial adjectives in Polish and English. *Linguistica Silesiana*, 28, 63–79.
- Chomsky, N. (2013). Problems of projection. *Lingua*, 130, 33–49. <https://doi.org/10.1016/j.lingua.2012.12.003>
- Collins, C. (2005). A smuggling approach to the passive in English. *Syntax*, 8(2), 81–120. <https://doi.org/10.1111/j.1467-9612.2005.00076.x>
- Coon, J., & Preminger, O. (2012). Towards a unified account of person splits. In J. Choi (Ed.), *Proceedings of the 29th West Coast Conference on Formal Linguistics* (pp. 310–8). Cascadia Press.
- D'Alessandro, R. (2017). When you have too many features: Auxiliaries, agreement and clitics in Italian varieties. *Glossa: A Journal of General Linguistics*, 2(1), 1–36. <https://doi.org/10.5334/gjgl.102>
- D'Alessandro, R., & Roberts, I. (2010). Past participle agreement in Abruzzese: Split auxiliary selection and the null-subject parameter. *Natural Language & Linguistic Theory*, 28(1), 41–72. <https://doi.org/10.1007/s11049-009-9085-1>
- De Belder, M. (2011). *Roots and affixes: Eliminating lexical categories from syntax* [Doctoral dissertation, Utrecht University]. <https://www.lotpublications.nl/roots-and-affixes-roots-and-affixes-eliminating-lexical-categories-from-syntax>
- Embick, D. (2004). On the structure of resultative participles in English. *Linguistic Inquiry*, 35(3), 355–92. <https://doi.org/10.1162/0024389041402634>
- Fanselow, G. (2004). The MLC and derivational economy. In A. Stepanov, G. Fanselow, & R. Vogel (Eds.), *Minimality effects in syntax* (pp. 73–124). De Gruyter Mouton. <https://doi.org/10.1515/9783110197365.73>
- Ferreira, F., Bailey, K. G. D., & Ferraro, V. (2002). Good-enough representations in language comprehension. *Current Directions in Psychological Science*, 11(1), 11–15. <https://www.jstor.org/stable/20182754>
- Frasson, A. (2025). The syntax of Serbian how-complements. *Languages*, 10(9), 227. <https://doi.org/10.3390/languages10090227>
- Frazier, L., & Rayner, K. (1982). Making and correcting errors during sentence comprehension: Eye movements in the analysis of structurally ambiguous sentences. *Cognitive Psychology*, 14(2), 178–210. [https://doi.org/10.1016/0010-0285\(82\)90008-1](https://doi.org/10.1016/0010-0285(82)90008-1)
- Gehrke, B. (2015). Adjectival participles, event kind modification and pseudo-incorporation. *Natural Language & Linguistic Theory*, 33(3), 897–938. <https://doi.org/10.1007/s11049-015-9295-7>

- Hale, J. (2001). A probabilistic Earley parser as a psycholinguistic model. In *Proceedings of the Second Meeting of the North American Chapter of the Association for Computational Linguistics on Language Technologies* (pp. 1–8). Association for Computational Linguistics.
- Halle, M., & Marantz, A. (1994). Some key features of Distributed Morphology. *MIT Working Papers in Linguistics*, 21, 275–88.
- Halpert, C. (2015). *Argument licensing and agreement*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780190256470.001.0001>
- Klemensiewicz, Z., Lehr-Splawiński, T., & Urbańczyk, S. (1985). *Gramatyka historyczna języka polskiego*. Państwowe Wydawnictwo Naukowe.
- Landau, I. (2010). *The locative syntax of experiencers*. MIT Press. <https://doi.org/10.7551/mitpress/8387.001.0001>
- Marín, R., & Fábregas, A. (2021). Participles and gerunds. In A. Fábregas, V. Acedo-Matellán, G. Armstrong, M. C. Cuervo, & I. Pujol Payet (Eds.), *The Routledge handbook of Spanish morphology* (pp. 484–98). Routledge. <https://doi.org/10.4324/9780429318191>
- Müller, G. (2001). Order preservation, parallel movement, and the emergence of the unmarked. In G. Legendre, J. Grimshaw, & S. Vikner (Eds.), *Optimality-theoretic syntax* (pp. 279–313). MIT Press. <https://doi.org/10.7551/mitpress/5161.003.0013>
- Partee, B. (1987). Noun phrase interpretation and type-shifting principles. In J. Groenendijk, D. De Jongh, & M. Stockhof (Eds.), *Studies in discourse representation theory and the theory of generalized quantifiers* (pp. 115–43). Foris Publications. <https://doi.org/10.1002/9780470758335.ch15>
- Pritchett, B. (1988). Garden path phenomena and the grammatical basis of language processing. *Language*, 64(3), 539–76. <https://doi.org/10.2307/414532>
- Sleeman, P. (2011). Verbal and adjectival participles: Position and internal structure. *Lingua*, 121(10), 1569–87. <https://doi.org/10.1016/j.lingua.2011.05.001>
- Svenonius, P. (1994). C-selection as feature-checking. *Studia Linguistica*, 48(2), 133–55. <https://doi.org/10.1111/j.1467-9582.1994.tb00853.x>
- Talsma, M. (2025). *Argument structure as a function of Merge: From UTAH to IDAHO*. [Doctoral dissertation, University of Groningen]. <https://research.rug.nl/en/publications/argument-structure-as-a-function-of-merge-from-utah-to-idaho/>
- Trueswell, J., Tanenhaus, M., & Kello, C. (1993). Verb-specific constraints in sentence processing: Separating effects of lexical preference from garden-paths. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 19(3), 528–53. <https://doi.org/10.1037/0278-7393.19.3.528>
- van der Wal, J. (2022). *A featural typology of Bantu agreement*. Oxford University Press. <https://doi.org/10.1093/oso/9780198844280.001.0001>
- Živojinović, J. (2021). *The development of the Latin gerund in Rhaeto-Romance* [Doctoral dissertation, University of Verona/The Arctic University of Norway]. https://iris.univr.it/retrieve/e14ff6e5-ca8c-0209-e053-6605fe0ad24c/Tesi_Zivojinovic.pdf
- Żychliński, S. (2011). A recalcitrant nature of object experiencers. In J. Bielak & M. Pawlak (Eds.), *New perspective in language, discourse and translation studies* (pp. 77–90). Springer. https://doi.org/10.1007/978-3-642-20083-0_6